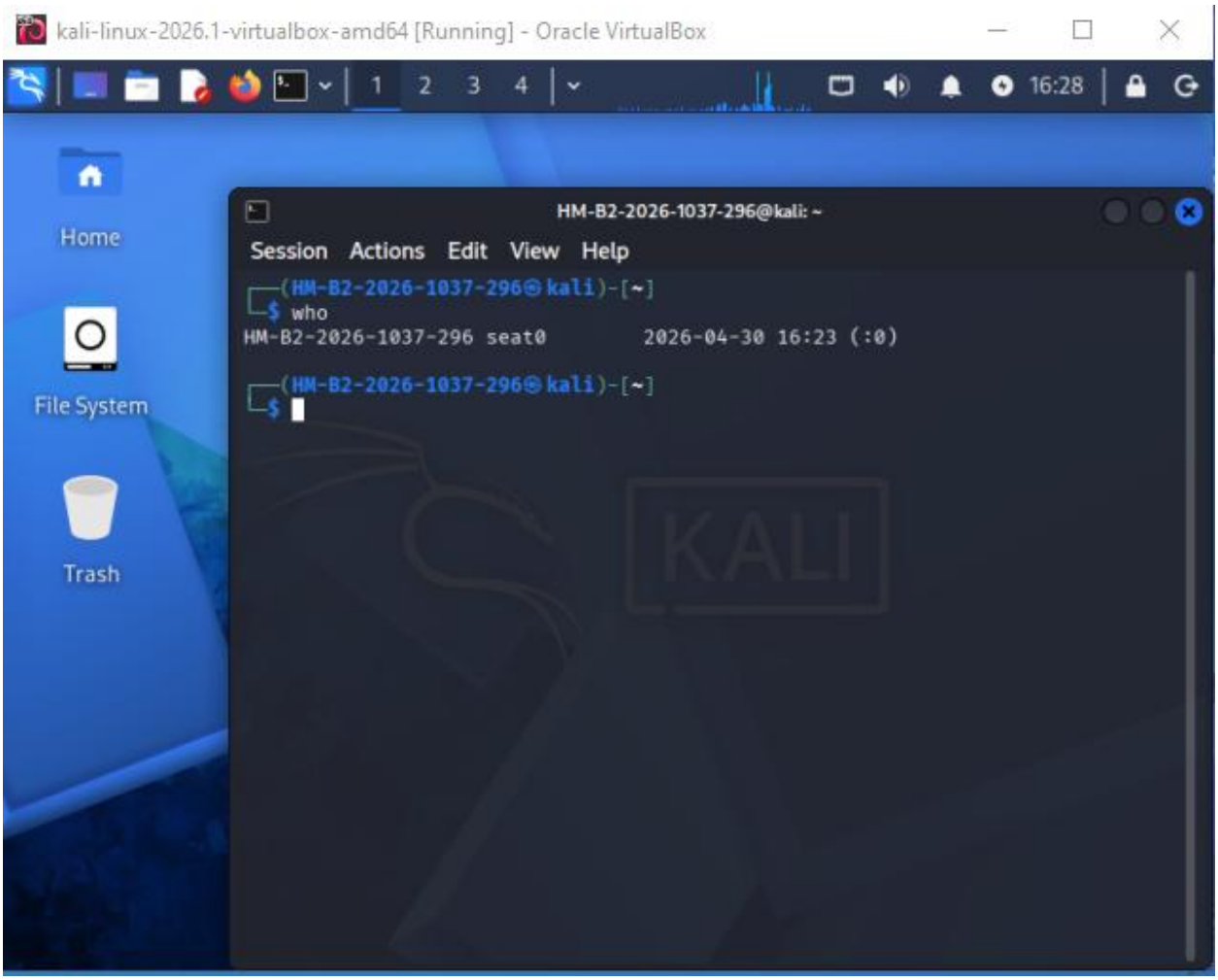
Syed Muhammad Abdullah Shah

HM-B2-2026-1037-296

Q1

We are creating new user for kali linux. By default, it was kali, kali but I created new user with my roll number of hunarmand.

After that I use who command in terminal for confirmation.



Q2

1. nslookup google.com

Telling Ip address of google

To verify if the DNS server is working correctly and translating website names into IP addresses

```
C:\Users\PC\HM-B2-2026-1037-296>nslookup google.com
Server:  one.one.one.one
Address:  1.1.1.1

Non-authoritative answer:
Name:    google.com
Addresses:  2404:6800:4009:810::200e
          142.250.70.46
```

2. systeminfo

Loads and displays information about my computer

To quickly check hardware specifications, OS build, and system uptime

```
C:\Users\PC\HM-B2-2026-1037-296>systeminfo

Host Name:                 DESKTOP-3RE1M6A
OS Name:                   Microsoft Windows 10 Pro
OS Version:                10.0.19045 N/A Build 19045
OS Manufacturer:           Microsoft Corporation
OS Configuration:          Standalone Workstation
OS Build Type:             Multiprocessor Free
```

3. tree

Graphically displays the folder structure of a drive or path

To get a quick, visual understanding of a complex folder structure

```
C:\Users\PC\HM-B2-2026-1037-296>tree
Folder PATH listing for volume Windows 10
Volume serial number is 5C8B-6451
C:.
No subfolders exist
```

4. getmac

Return mac address of networks protocol associated with each address of all networkd cards.

To find the physical address of the network card

```
C:\Users\PC\HM-B2-2026-1037-296>getmac

Physical Address    Transport Name
=================== ============================================================
N/A                 Hardware not present
38-2C-4A-C5-7B-69   \Device\Tcpip_{4049A423-6EBA-4F2D-8216-99C7C93F5198}
00-FF-04-87-AD-F2   Media disconnected
0A-00-27-00-00-08   \Device\Tcpip_{3B8D9B46-CE8C-44F7-8968-F59600D0DF4B}
```

5. whoami

Displays the domain and username

To confirm which user account is using the cmd

```
C:\Users\PC\HM-B2-2026-1037-296>whoami
desktop-3re1m6a\pc
```

6. hostname

Displays the current name of your computer

To identify the exact network name of the machine

```
C:\Users\PC\HM-B2-2026-1037-296>hostname
DESKTOP-3RE1M6A
```

7. ver

Displays the exact Windows version you are running

To check the specific Windows build number

```
C:\Users\PC\HM-B2-2026-1037-296>ver

Microsoft Windows [Version 10.0.19045.6466]
```

8. netstat

Displays active TCP connections, ports

To monitor active network connections

```
C:\Users\PC\HM-B2-2026-1037-296>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    192.168.18.174:49679   4.213.25.242:https     ESTABLISHED
  TCP    192.168.18.174:49736   lcbomp-in-f188:5228    ESTABLISHED
  TCP    192.168.18.174:50289   13.107.213.48:https    CLOSE_WAIT
  TCP    192.168.18.174:50291   server:6196            ESTABLISHED
  TCP    192.168.18.174:50846   52.111.231.79:https    ESTABLISHED
  TCP    192.168.18.174:50851   13.107.226.254:https   CLOSE_WAIT
  TCP    192.168.18.174:50990   239:https              ESTABLISHED
```

9.tracert google.com

Traces the route that packets take to reach a destination network

To pinpoint exactly where a network delay or failure is occurring

```
C:\Users\PC\HM-B2-2026-1037-296>tracert google.com

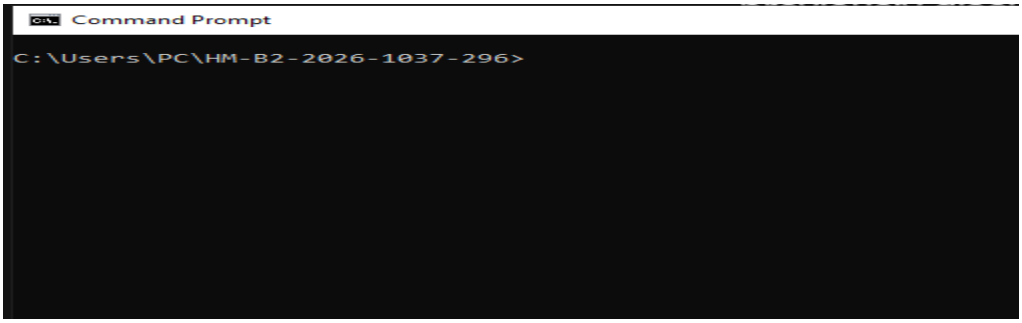
Tracing route to google.com [142.251.222.110]
over a maximum of 30 hops:

  1     1 ms    <1 ms    <1 ms  192.168.18.1
  2     3 ms     2 ms     2 ms  0.e1-1-3-1.cr01-lhr.cyber.net.pk [202.163.12
  3     4 ms     3 ms     3 ms
```

10. cls

Clears the command prompt screen of all previous output.

To clean up a cluttered terminal screen



Q3

1. pwd

It tells you exactly which folder you are currently inside

To verify my current location in directory

2.uname -a

Prints basic information about the operating system

To check the exact kernel version and system architecture

3. df -h

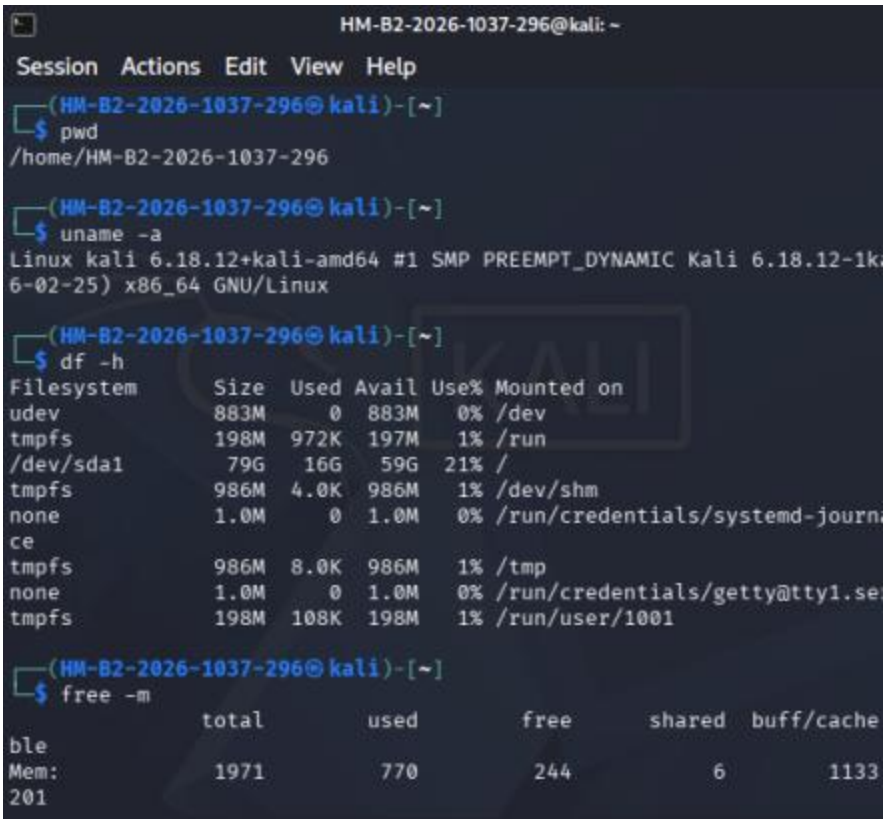
Reports on the amount of available disk space being used by file systems

To quickly check how much storage space is available

4.free -m

Displays the total amount of free and used physical memory

To monitor RAM and swap usage



5. history

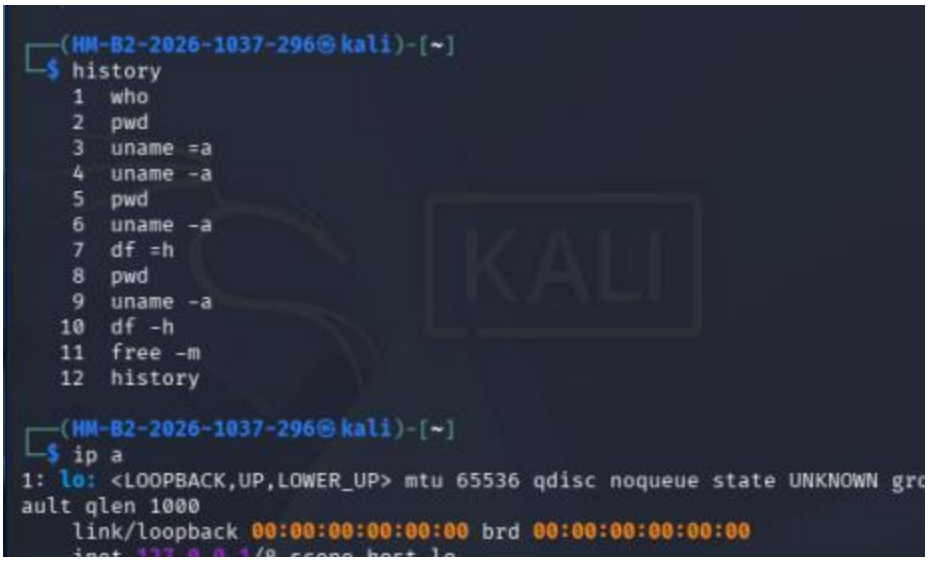
Shows a numbered list of all the previous commands in terminal

To easily find and re-run a long or complex command I typed earlier

6.ip a

Displays IP addresses and property information

To find my local IP address and network interface details



7. top

Provides a dynamic, real-time view of a running system

To identify which specific processes or applications are consuming the most CPU



8. find . -name "test.txt"

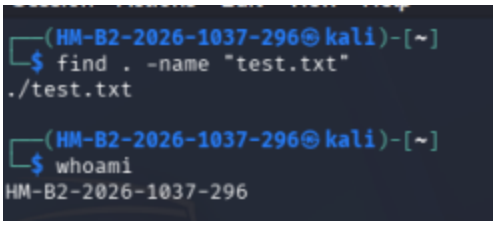
Searches the directory tree for file

To quickly locate a specific file

9. whoami

Tells about you

To confirm I am operating under the correct user account



10.clear

Clear terminal

To wipe the screen of past command outputs

